

Seaborn Assingment

1.

Seaborn Assignment

+ Code + Text

1.install seaborn and import it

```
pip install seaborn
```

```
Requirement already satisfied: seaborn in /usr/local/lib/python3.12/dist-packages (0.13.2)
Requirement already satisfied: numpy!=1.24.0,>=1.20 in /usr/local/lib/python3.12/dist-packages (from seaborn) (2.0.2)
Requirement already satisfied: pandas>=1.2 in /usr/local/lib/python3.12/dist-packages (from seaborn) (2.2.2)
Requirement already satisfied: matplotlib=3.6.1,>=3.4 in /usr/local/lib/python3.12/dist-packages (from seaborn) (3.10.0)
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib=3.6.1,>=3.4->seaborn) (1.3.3)
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib=3.6.1,>=3.4->seaborn) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib=3.6.1,>=3.4->seaborn) (4.63.0)
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib=3.6.1,>=3.4->seaborn) (1.5.0)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib=3.6.1,>=3.4->seaborn) (26.2)
Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-packages (from matplotlib=3.6.1,>=3.4->seaborn) (11.3.6)
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib=3.6.1,>=3.4->seaborn) (3.3.2)
Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.12/dist-packages (from matplotlib=3.6.1,>=3.4->seaborn) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas>=1.2->seaborn) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas>=1.2->seaborn) (2026.3)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.7->matplotlib=3.6.1,>=3.4->seaborn) (1.17.0)
```

2&3.

2.load the iris dataset

3.display data inspection

```
import seaborn as sns
import matplotlib.pyplot as plt
import pandas as pf
df = sns.load_dataset("iris")
print("5 first rows")
print(df.head())
print("5 last rows")
print(df.tail())
print(df.shape)
print(df.columns)
print(df.dtypes)
print(df.info())
print(df.describe())
print(df.isnull().sum())
print(df.duplicated().sum())
print(df.nunique())
print(df["species"].value_counts())
```

```

*** 5 first rows
   sepal_length sepal_width petal_length petal_width species
0          5.1         3.5         1.4         0.2 setosa
1          4.9         3.0         1.4         0.2 setosa
2          4.7         3.2         1.3         0.2 setosa
3          4.6         3.1         1.5         0.2 setosa
4          5.0         3.6         1.4         0.2 setosa
5 last rows
   sepal_length sepal_width petal_length petal_width species
145          6.7         3.0         5.2         2.3 virginica
146          6.3         2.5         5.0         1.9 virginica
147          6.5         3.0         5.2         2.0 virginica
148          6.2         3.4         5.4         2.3 virginica
149          5.9         3.0         5.1         1.8 virginica
(150, 5)
Index(['sepal_length', 'sepal_width', 'petal_length', 'petal_width',
       'species'],
      dtype='object')
sepal_length    float64
sepal_width     float64
petal_length    float64
petal_width     float64
species         object
dtype: object
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
#   Column          Non-Null Count  Dtype
---  ---
0   sepal_length    150 non-null   float64
1   sepal_width     150 non-null   float64
2   petal_length    150 non-null   float64

```

s  Terminal

```

0   sepal_length    150 non-null   float64
1   sepal_width     150 non-null   float64
2   petal_length    150 non-null   float64
3   petal_width     150 non-null   float64
4   species         150 non-null   object
dtypes: float64(4), object(1)
memory usage: 6.0+ KB
None
      sepal_length sepal_width petal_length petal_width
count    150.000000    150.000000    150.000000    150.000000
mean       5.843333       3.057333       3.758000       1.199333
std        0.828066       0.435866       1.765298       0.762238
min        4.300000       2.000000       1.000000       0.100000
25%        5.100000       2.800000       1.600000       0.300000
50%        5.800000       3.000000       4.350000       1.300000
75%        6.400000       3.300000       5.100000       1.800000
max        7.900000       4.400000       6.900000       2.500000
sepal_length    0
sepal_width     0
petal_length    0
petal_width     0
species         0
dtype: int64
1
sepal_length    35
sepal_width     23
petal_length    43
petal_width     22
species         3
dtype: int64
species
setosa         50
versicolor    50
virginica      50
..            ..

```

variables  Terminal

```

count      150.000000    150.000000    150.000000    150.000000
mean        5.843333     3.057333     3.758000     1.199333
std         0.828066     0.435866     1.765298     0.762238
min         4.300000     2.000000     1.000000     0.100000
25%         5.100000     2.800000     1.600000     0.300000
50%         5.800000     3.000000     4.350000     1.300000
75%         6.400000     3.300000     5.100000     1.800000
max         7.900000     4.400000     6.900000     2.500000
sepal_length  0
sepal_width   0
petal_length  0
petal_width   0
species       0
dtype: int64
1
sepal_length  35
sepal_width   23
petal_length  43
petal_width   22
species       3
dtype: int64
species
setosa        50
versicolor   50
virginica     50
Name: count, dtype: int64

```

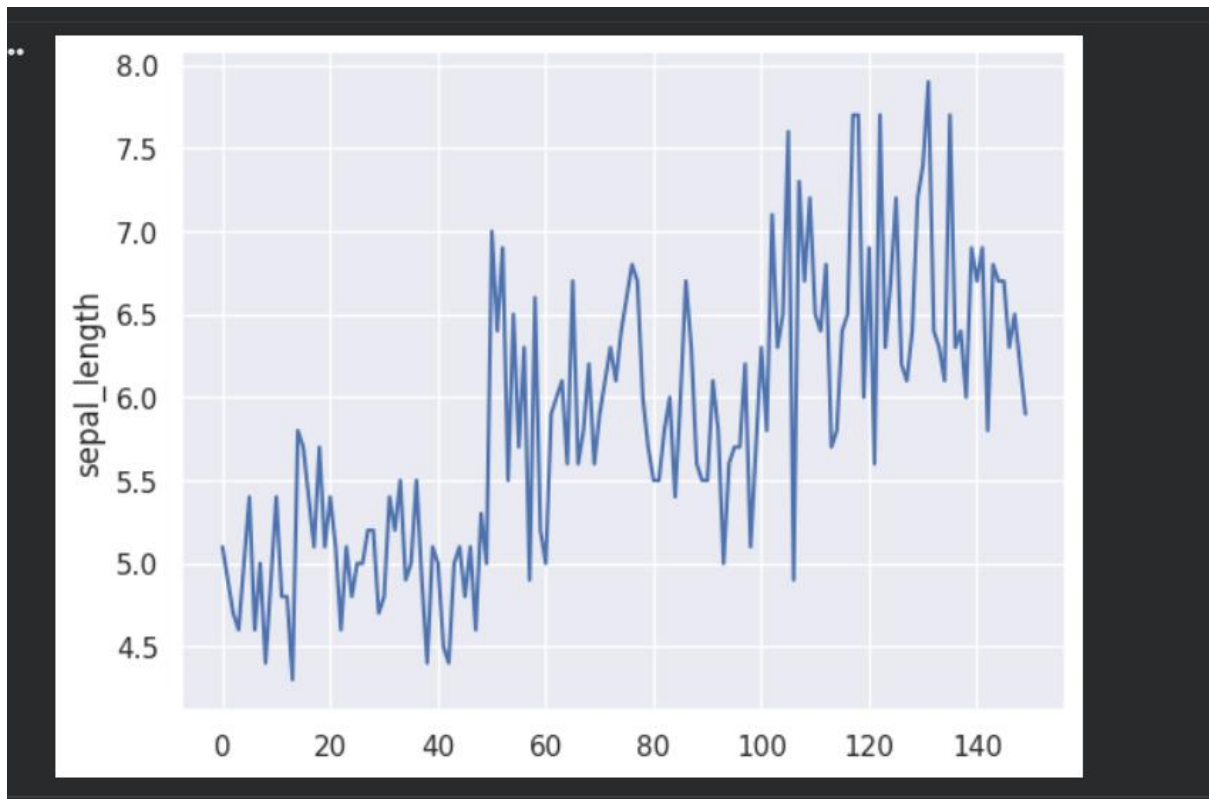
4.

4.apply different seaborn themes and observe the change

```

import seaborn as sns
import matplotlib.pyplot as plt
import pandas as pf
df = sns.load_dataset("iris")
sns.set_theme(style="darkgrid")
sns.lineplot(df["sepal_length"])
plt.show()

```



5&6.

5.load another dataset,

6.create a lineplot and show any two columns plotted

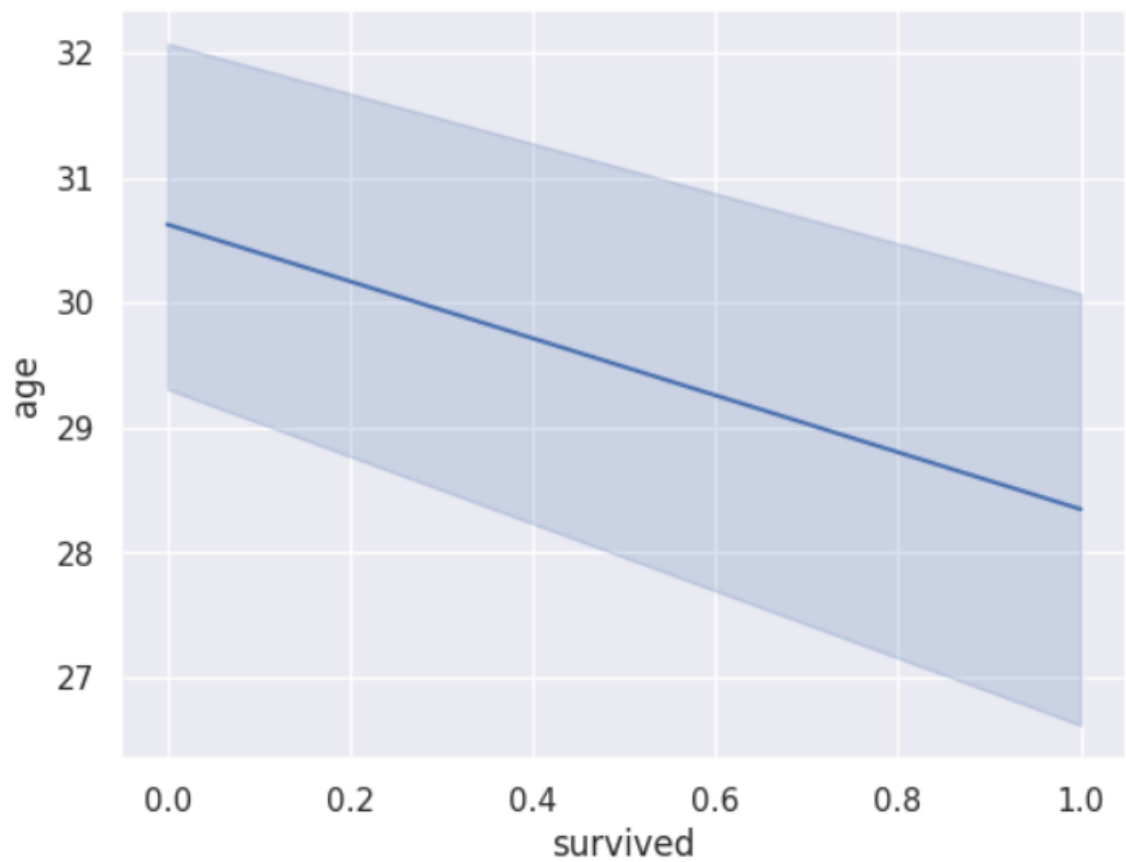
```
import seaborn as sns
import matplotlib.pyplot as plt
import pandas as pf
df = sns.load_dataset("titanic")
print(df.head())
sns.lineplot(x=df["survived"],y=df["age"])
plt.show()
```

```

...   survived  pclass   sex   age  sibsp  parch   fare embarked  class \
0      0        3    male  22.0     1      0    7.2500      S   Third
1      1        1    female 38.0     1      0   71.2833      C   First
2      1        3    female 26.0     0      0    7.9250      S   Third
3      1        1    female 35.0     1      0   53.1000      S   First
4      0        3    male  35.0     0      0    8.0500      S   Third

      who  adult_male  deck  embark_town  alive  alone
0    man           True  NaN  Southampton  no   False
1  woman          False   C   Cherbourg  yes   False
2  woman          False  NaN  Southampton  yes    True
3  woman          False   C   Southampton  yes   False
4    man           True  NaN  Southampton  no    True

```



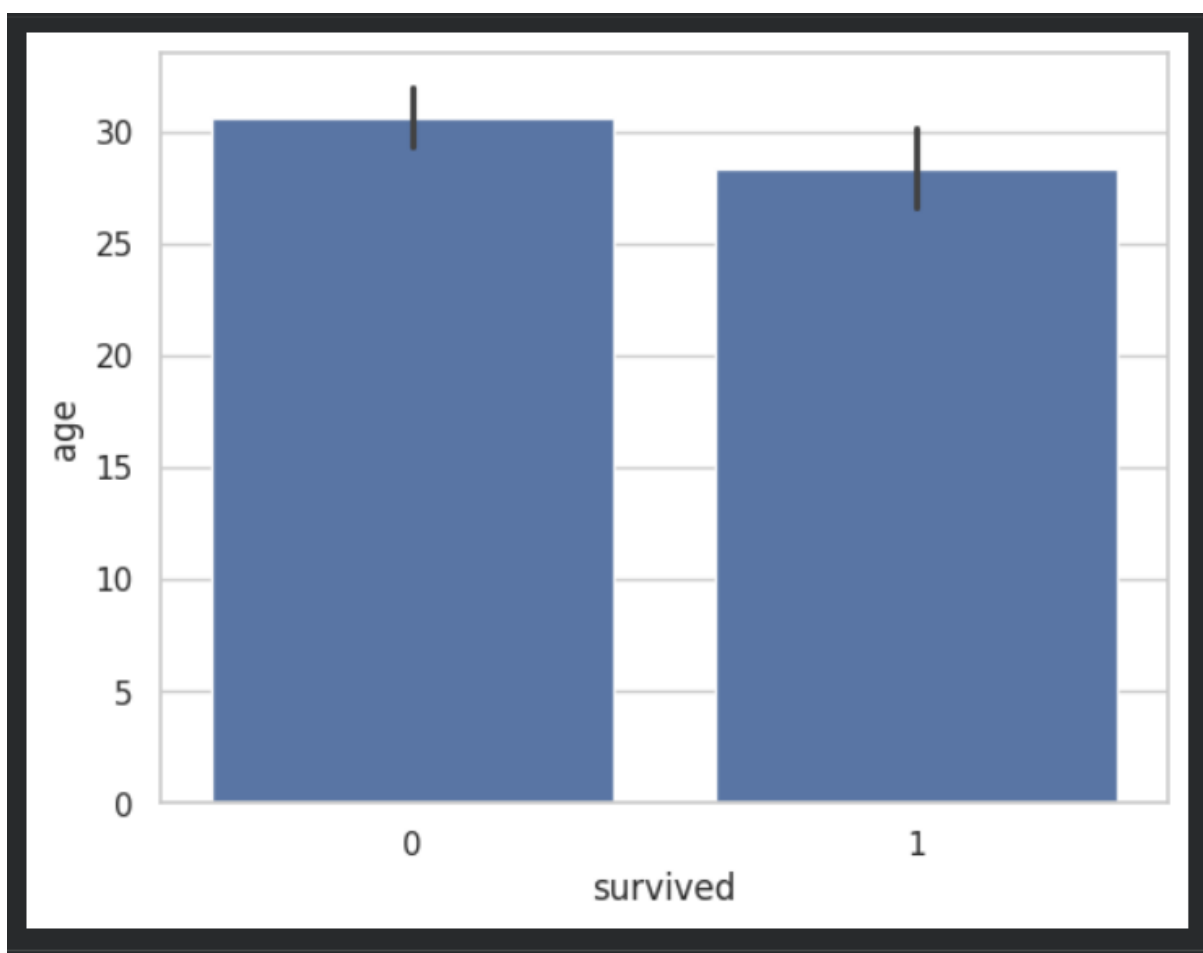
7.

7.create a barchart

```

▶ sns.set_theme(style="whitegrid")
  sns.barplot(data=df,x="survived",y="age")
  plt.show()

```

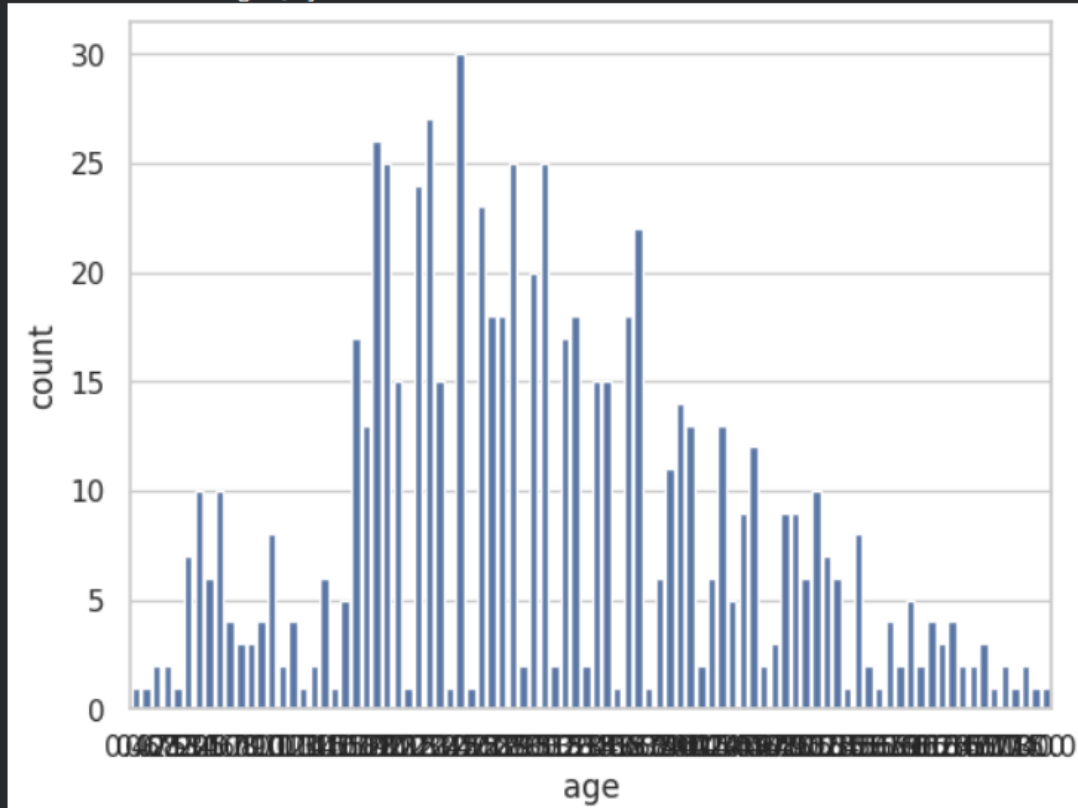


8.

8.create a countplot

```
sns.countplot(data=df,x="age")
```

```
<Axes: xlabel='age', ylabel='count'>
```



9&10&11&12.

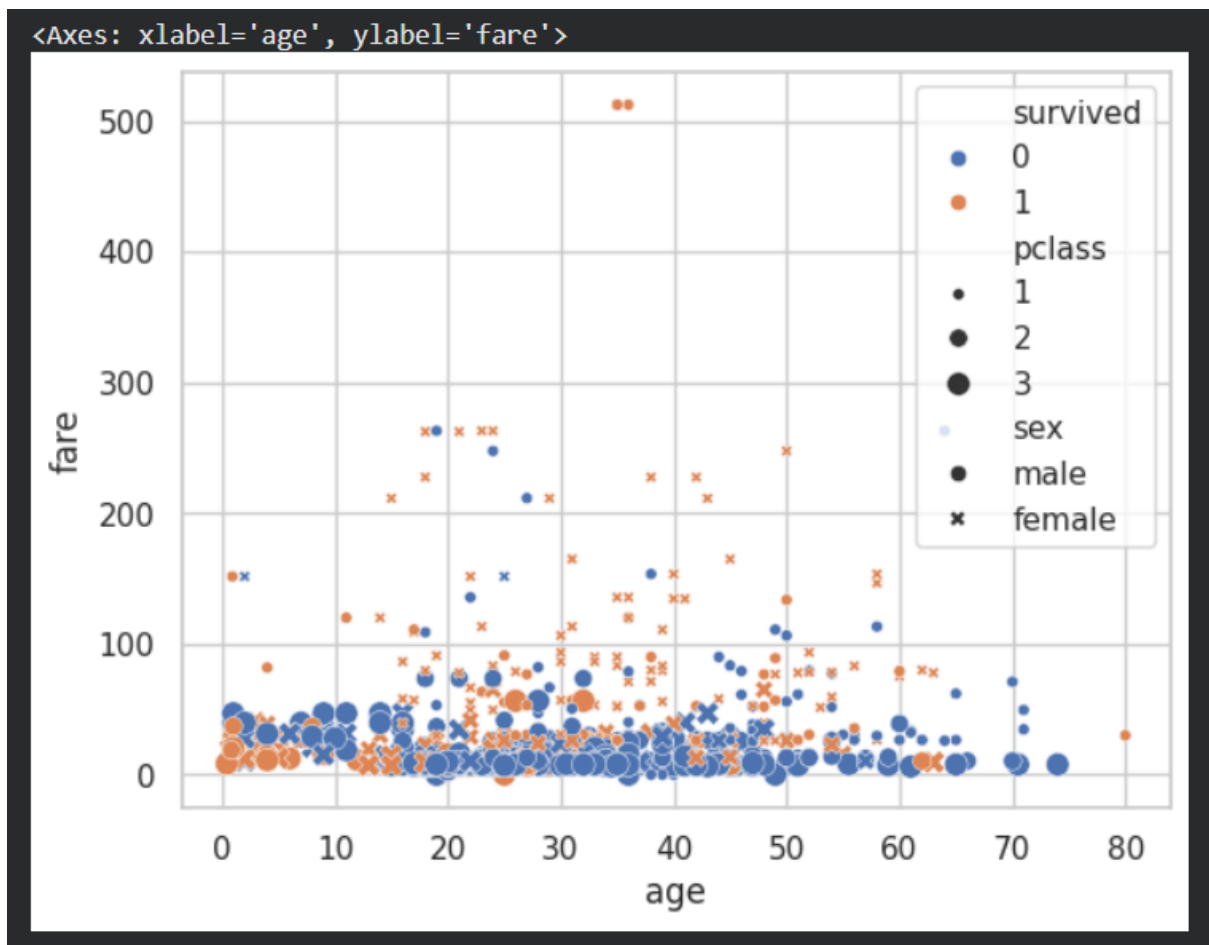
9.create a scatterplot(vs)

10.hue

11.style

12.size

```
sns.scatterplot(data=df,x="age",y="fare",hue="survived",style="sex",size="pclass")
```



13&14&15.

13.load any dataset

14.create a histogram for a column

15.create a histogram with bins

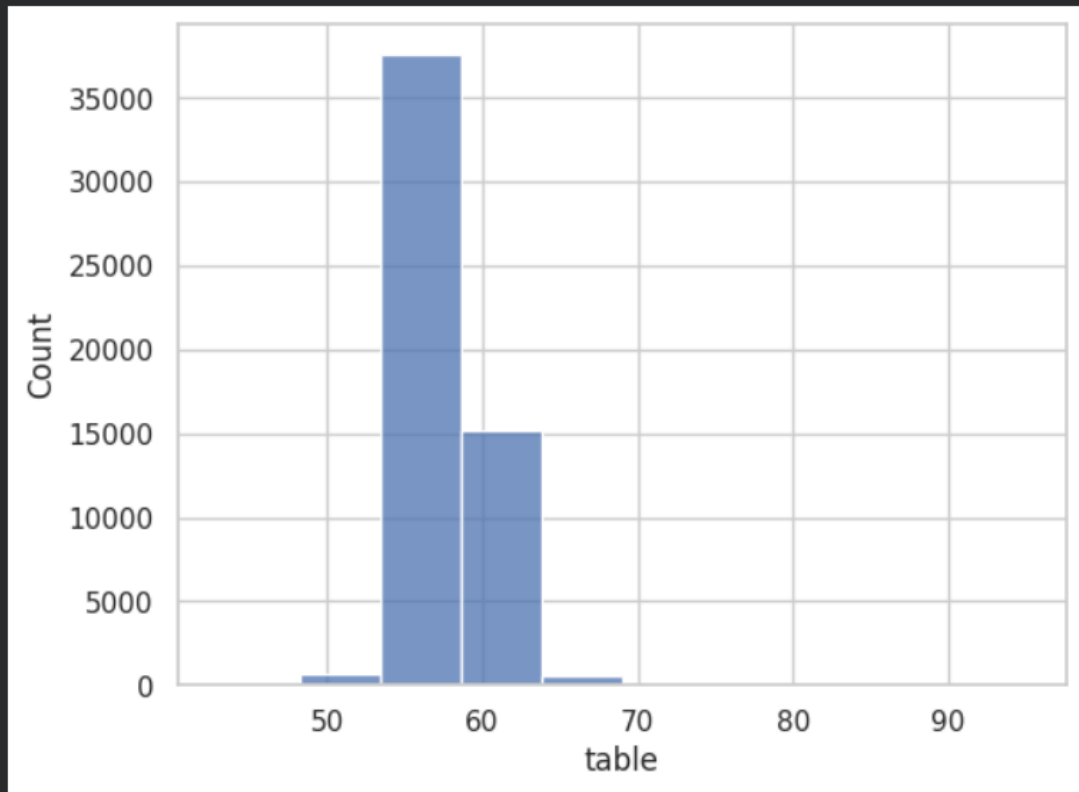
```
6] 0s ▶ import seaborn as sns
import matplotlib.pyplot as plt
import pandas as pd

df=sns.load_dataset("diamonds")
print(df.head())
sns.histplot(data=df,x="table",bins=10)
plt.show()
```

```

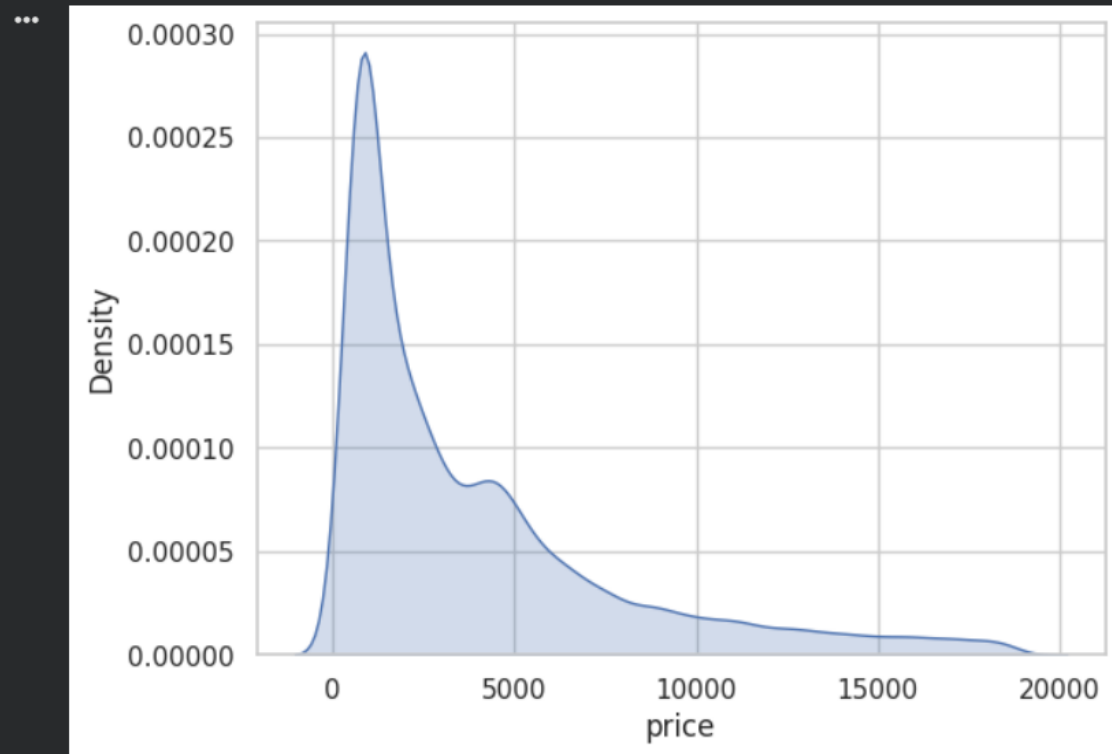
...   carat    cut color clarity depth  table  price     x     y     z
0   0.23  Ideal     E    SI2   61.5   55.0   326  3.95  3.98  2.43
1   0.21 Premium     E    SI1   59.8   61.0   326  3.89  3.84  2.31
2   0.23   Good     E    VS1   56.9   65.0   327  4.05  4.07  2.31
3   0.29 Premium     I    VS2   62.4   58.0   334  4.20  4.23  2.63
4   0.31   Good     J    SI2   63.3   58.0   335  4.34  4.35  2.75

```



16.

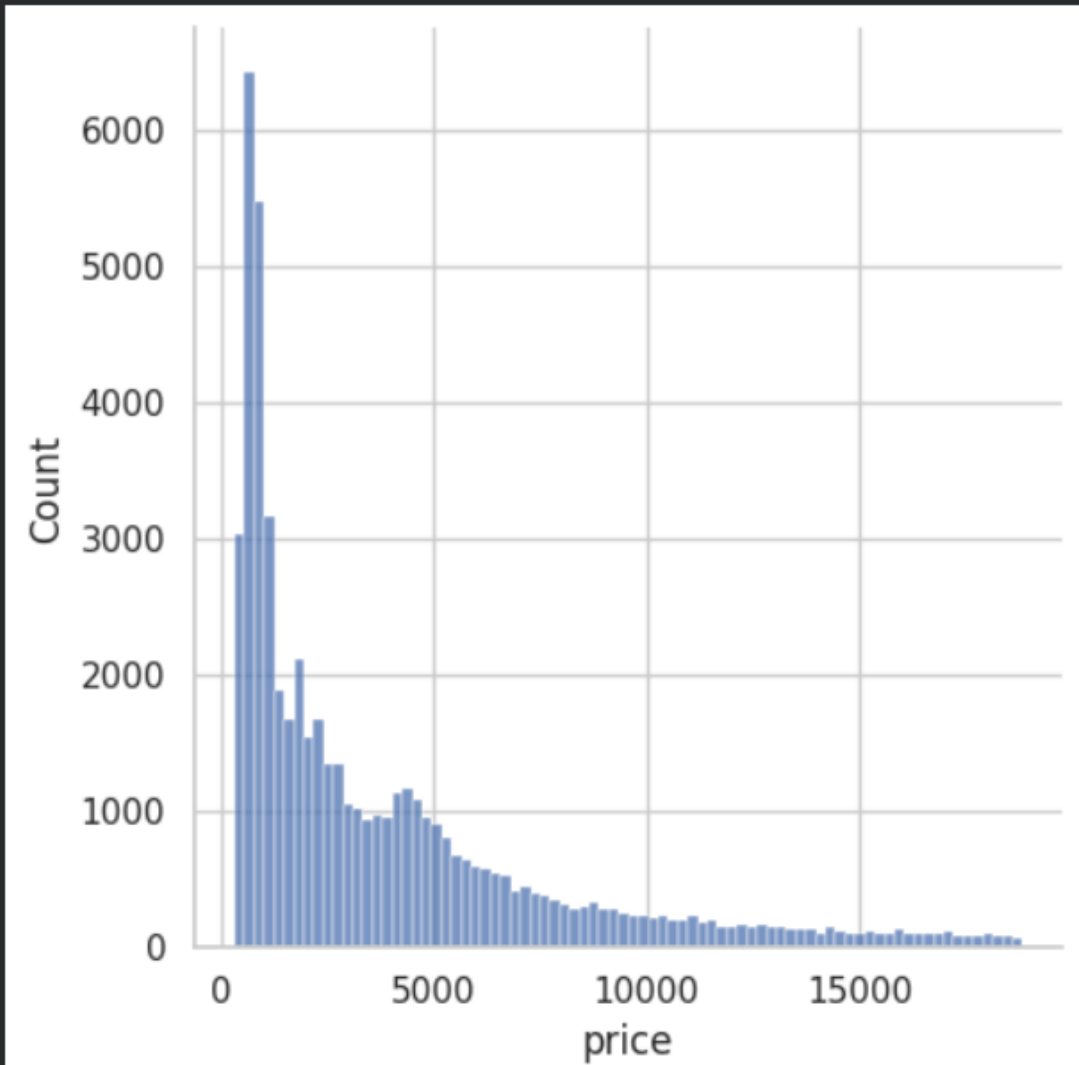
```
sns.kdeplot(data=df,x="price",fill=True)  
plt.show()
```



17.

```
sns.displot(data=df,x="price")  
plt.show()
```

...



18&19.

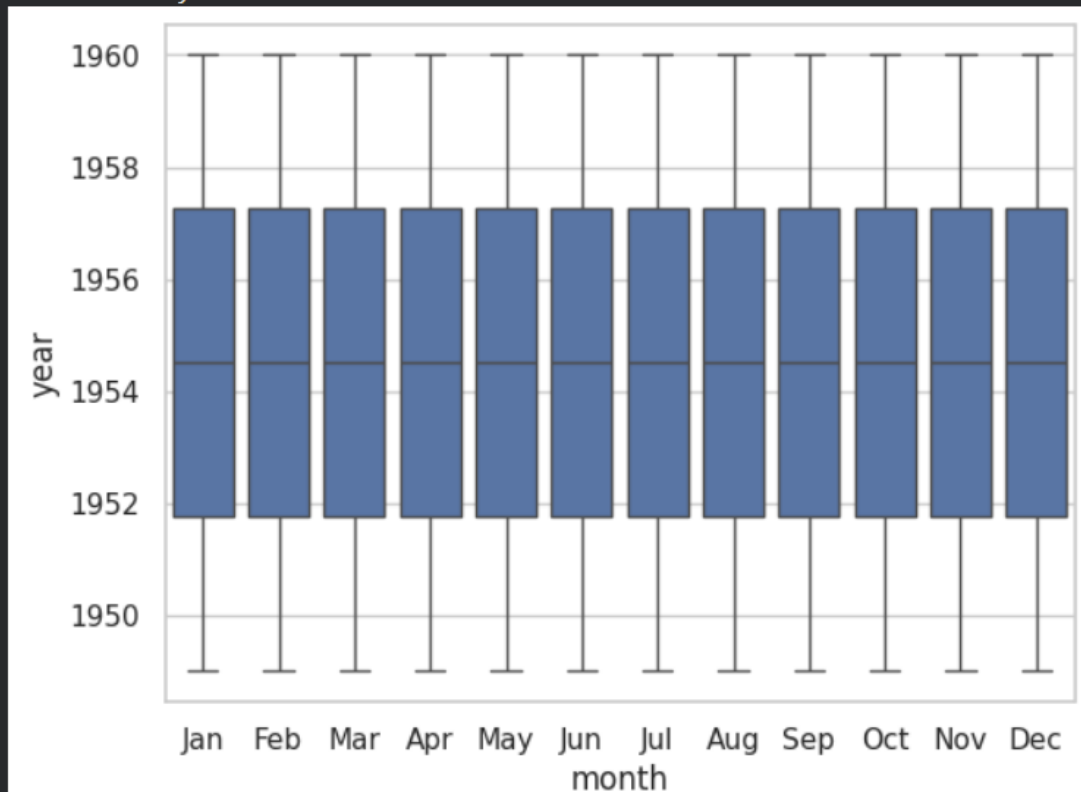
18.load any dataset

19.create a boxplot with 2 columns(vs)

```
import seaborn as sns
import matplotlib.pyplot as plt
import pandas as pd

df=sns.load_dataset("flights")
print(df.head())
sns.boxplot(data=df,x="month",y="year")
plt.show()
```

```
...   year month  passengers
0   1949   Jan         112
1   1949   Feb         118
2   1949   Mar         132
3   1949   Apr         129
4   1949   May         121
```

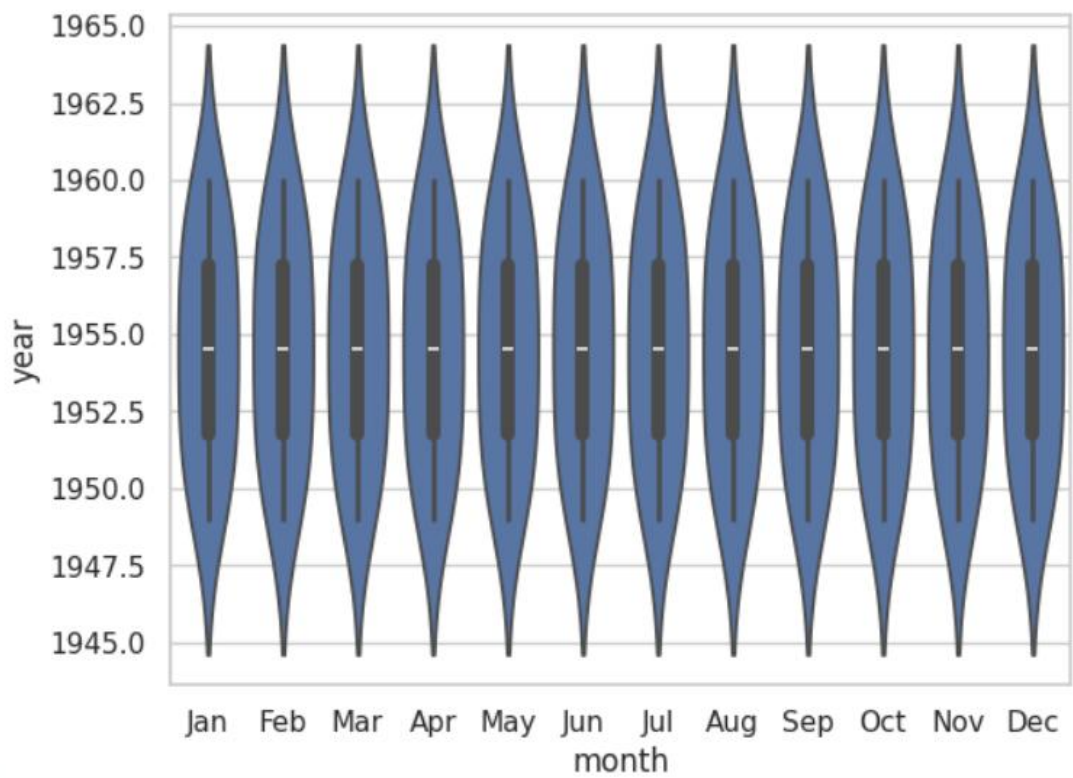


20.

20.create a violin plot

```
sns.violinplot(data=df,x="month",y="year")  
plt.show()
```

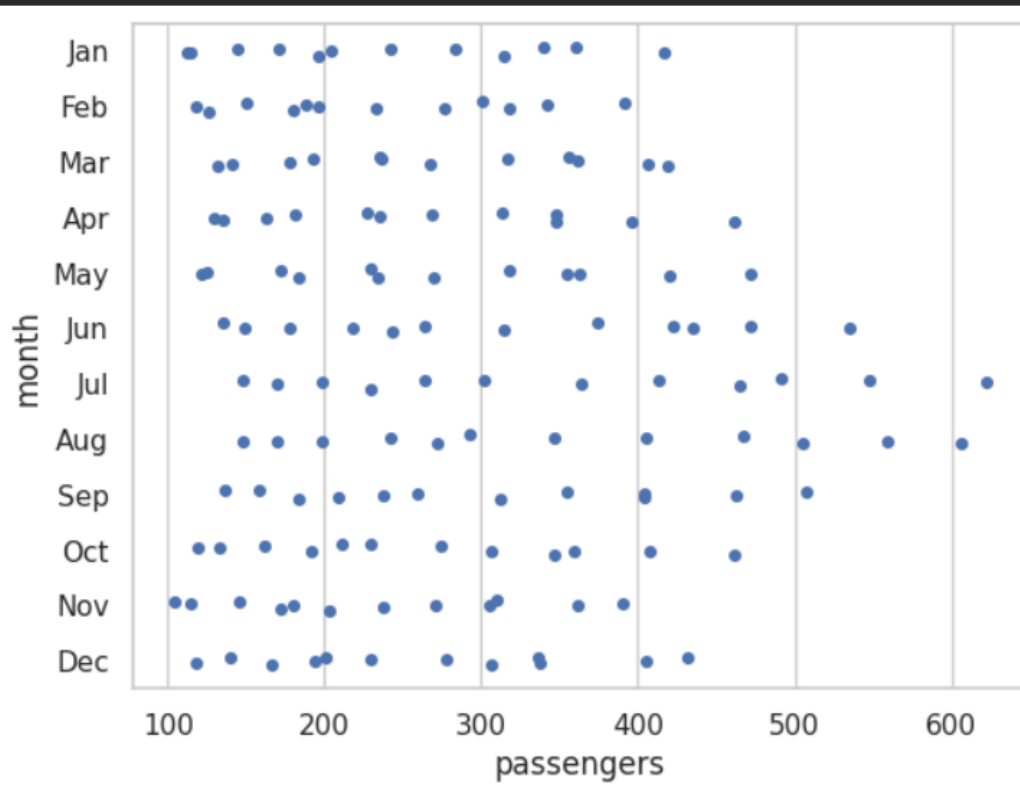
...



21.

21.create a strip plot with jitter

```
sns.stripplot(data=df,x="passengers",y="month",jitter=True)  
plt.show()
```

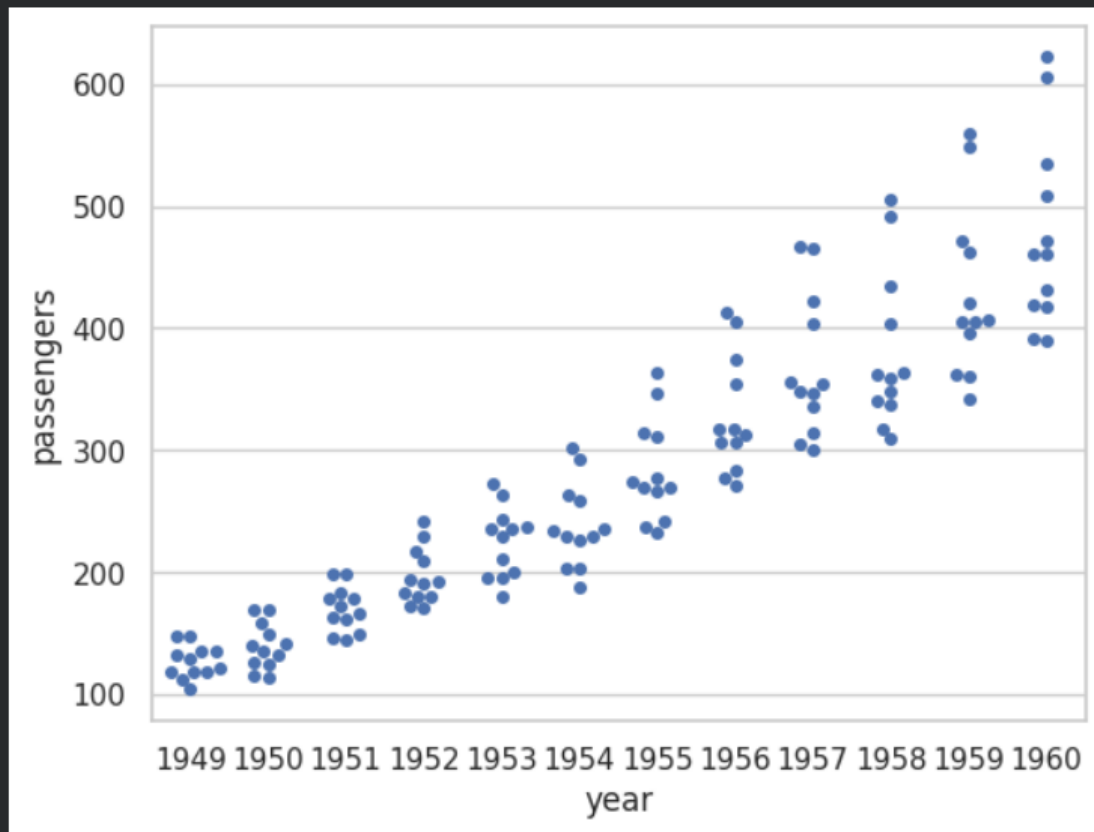


22.

22.create a swarm plot

+ Co

```
sns.swarmplot(data=df,x="year",y="passengers")  
plt.show()
```



23.

23.create a boxplot using hue

```
import seaborn as sns  
import matplotlib.pyplot as plt  
import pandas as pd  
df =sns.load_dataset("iris")  
print(df.head())  
sns.boxplot(data=df,x="petal_length",y="sepal_width",hue="species")  
plt.show()
```

```
...
  sepal_length sepal_width petal_length petal_width species
0          5.1          3.5          1.4          0.2  setosa
1          4.9          3.0          1.4          0.2  setosa
2          4.7          3.2          1.3          0.2  setosa
3          4.6          3.1          1.5          0.2  setosa
4          5.0          3.6          1.4          0.2  setosa
```

